# HEFLGuard: Backdoor Detection in Homomorphic Encryption-Based Federated Learning

Congyi Li, Peizhuo Lv, Jinwen He, Kai Chen, *Member, IEEE*, and Hong Zhu

***Abstract*—Homomorphic encryption-based federated learning (HEFL) strengthens privacy by aggregating encrypted model updates, but it also renders existing backdoor defenses that assume plaintext updates inapplicable. We present HEFLGuard, a single-server backdoor detection framework for HEFL in which the server constructs overlapping validation models from encrypted client groups and clients locally compare logits of the global and validation models on benign samples to expose backdoor behavior. HEFLGuard further combines consistency verification across non-IID validation groups with Byzantine fault-tolerant aggregation of client reports, ensuring robustness under heterogeneous data and Byzantine participants. We evaluate HEFLGuard on seven vision/text benchmarks under three backdoor types across IID and non-IID settings. HEFLGuard consistently reduces ASR from near 100% to nearly the no-backdoor level while keeping the drop in clean accuracy within 2.5%. Compared with prior work, HEFLGuard achieves higher robustness and deployability.**

***Index Terms*—Federated learning, backdoor detection, homomorphic encryption.**

## I. Introduction

FEDERATED learning (FL) is a widely used machine learning framework where multiple decentralized clients collaboratively train a model under the coordination of a central server without sharing their raw data [1]. The FL framework is designed to protect sensitive data from unauthorized access, so organizations often choose it when they want to keep their data private without sharing it with third parties. However, traditional FL still poses a risk of privacy leakage, as the model updates may reveal information about the local data or models [2]. For instance, adversaries may launch reconstruction [3] and inference attacks [4] to infer the localized data from the model updates. To address these challenges, homomorphic encryption (HE) is a promising and mainstream technique that allows computations on encrypted model updates [5]. In HE-based federated learning (HEFL) systems, clients encrypt their model updates with the private key before sending them to the central server, and the central server aggregates the encrypted updates without decrypting them [6]. In this way, the central server cannot access the raw model updates, and thus the privacy of the local data and models is preserved.

Federated learning systems are vulnerable to backdoor attacks. The backdoor attacks in FL [7], [8] refer to the attackers modifying the model uploaded by clients to the central server in various ways, such as adding a trigger pattern or changing the model parameters. A backdoor makes the global model misclassify triggered samples regardless of their actual content, while working normally otherwise. In critical fields like healthcare and finance, this could lead to misdiagnoses or financial fraud [9]. For traditional FL, the mainstream of defense methods is based on the similarity of the gradients from clients. Some use metrics such as cosine similarity, Euclidean distance, Pearson correlation coefficient, or K-means [10], [11], [12] to measure the gradient similarity. However, they all have a major drawback: they require the server to access the plaintext model parameters from the clients. For HEFL systems, these methods violate privacy-preserving principles and are unsuitable since the model parameters are encrypted and only accessible by the clients.

Recent research has explored backdoor defense in homomorphic encryption-based FL. Existing approaches primarily fall into two categories: based on multi-server collaboration and based on Zero-Knowledge Proofs (ZKP). The former, such as ShieldFL [13] and the method proposed by Yazdinejad et al. [14], rely on either multiple non-colluding servers or trusted third-party auditors for encrypted gradient analysis. The ZKP category, including EIFFeL [15], RoFL [16], and ACORN [17], employs zero-knowledge proofs to validate model updates against predefined constraints. However, these methods suffer from substantial computational overhead and lack effective mechanisms for handling Non-IID scenarios.

These approaches leave three critical gaps for practical HEFL deployments: (i) they often assume multiple non-colluding servers or trusted auditors, which contradicts the widely adopted single-server architecture in real-world systems; (ii) ZKP-based solutions incur substantial computational and communication overhead, making them difficult to scale to large client populations or resource-constrained devices; (iii) existing designs offer limited support for realistic challenges such as highly Non-IID data, colluding or Byzantine clients, and diverse backdoor strategies. To the best of our knowledge, there is no existing method that simultaneously

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preserves end-to-end encryption, avoids extra trusted parties, remains efficient, and robustly detects backdoors under these conditions in a standard single-server HEFL setting.

To address this challenge, we propose HEFLGuard, a pioneering backdoor detection scheme that works in homomorphic encryption-based federated learning systems. HEFLGuard employs an innovative architecture that facilitates systematic collaboration between the central server and clients, eliminating the need for additional servers or trusted parties. It uniquely leverages the comparison of model prediction logits from different models to measure consistency. To enable this comparison, the central server divides clients into groups, aggregates each group's models using HE to generate a validation model, and then randomly distributes the validation model along with the global model to clients. To enable this comparison, the central server divides clients into groups, aggregates each group's models using HE to generate a validation model, and then randomly distributes a validation model along with the global model to clients. Clients run the models on their local test samples, compare the logits values, and report the results to the central server. Finally, the central server conducts a cross-analysis of the detection results from clients, diagnoses whether the global model is backdoored and identifies malicious clients. Furthermore, we propose two enhancement schemes: the consistency verification that improves the robustness of HEFLGuard in Non-IID scenarios by identifying the consistent backdoor among the results of groups; and the Byzantine fault-tolerance verification that enables HEFLGuard to obtain accurate detection results despite the dishonest reports of the malicious client by using majority voting. They make HEFLGuard more resilient to various attacks and challenges in realistic settings.

We evaluate HEFLGuard on seven vision and text benchmarks: MNIST, FASHION, CHMNIST (colorectal cancer tissues), CIFAR10, CIFAR100, AG News, and Newsgroups, with three representative backdoor types under both IID and Non-IID settings. HEFLGuard consistently reduces the backdoor attack success rate from near 100% down to the no-backdoor level, while maintaining the drop in clean accuracy within 2.5%. Across these settings, HEFLGuard's malicious-client identification achieves near-100% true-positive rate with only a few percent false positives. Even on the most challenging CIFAR100 Non-IID case, the few falsely flagged clients have negligible impact on the global model. Our ablations motivate key design choices: analysis of the validation group size $k$ and attacker scaling factor $\lambda$ identifies robust operating points, while experiments with a classifier-only design propose a better efficiency–robustness trade-off. Under stronger adversaries with up to 30–40% malicious clients, including distributed backdoor attacks and cheating clients that report corrupted diagnostics, HEFLGuard still identifies the majority of adversaries while keeping the global backdoor success rate low. Finally, we compare HEFLGuard with the existing work Trim-mean [18], Krum [19], FLTrust [8], and RoFL [16], and find that HEFLGuard is more robust and practical.

Our main contributions are summarized as follows:

- We propose HEFLGuard, a backdoor detection framework specifically tailored for single-server HEFL, which operates without accessing plaintext updates, without additional trusted parties, and remains fully compatible with standard homomorphic encryption workflows.
- We propose two enhancement schemes for HEFLGuard: the consistency verification that enables it to work in both IID and Non-IID scenarios, and the Byzantine fault-tolerance verification that enables it to maintain accuracy despite the dishonesty of the malicious clients.
- We comprehensively evaluate HEFLGuard on seven real-world datasets. The results demonstrate its high performance, Byzantine-robustness, and low overhead.

## II. Background and Related Work

### A. Homomorphic Encryption-Based FL

Ensuring data privacy is crucial in federated learning, where sensitive information can be inferred from the raw model updates. Homomorphic encryption is a fundamental technique that addresses this challenge by enabling computations to be performed directly on encrypted data, without the need for decryption during the computation process [5], [20]. The central server aggregates the encrypted models, leveraging the additive homomorphic property of the cryptosystem. This property allows certain operations, like addition, to be performed on ciphertexts (encrypted data), generating an encrypted result that matches the operation performed on the corresponding plaintexts (unencrypted data). Many cryptosystems possess this additive homomorphic property, such as: RSA [21], Paillier [22], lattice-based [23], and ElGamal [24]. By preventing model leakage and tampering, HE-based FL finds widespread application in highly privacy-sensitive domains like healthcare and finance [9].

A typical homomorphic encryption-based FL system operates as follows [6], [23]. First, an asymmetric cryptography key pair is generated: a public key $pk$ for encryption and a private key $sk$ for decryption. Clients hold both $pk$ and $sk$, while the server possesses only $pk$. During aggregation, each client encrypts their local model updates $M_i$ using $pk$ to obtain ciphertexts $E_{pk}(M_i)$, which are sent to the server. The server directly aggregates these ciphertexts to obtain the encrypted global model $E_{pk}(M_g)$, which is then distributed to clients. Clients decrypt $E_{pk}(M_g)$ using $sk$ to retrieve the global model parameters $D_{sk}(E_{pk}(M_g))$. Throughout this process, the server cannot access any unencrypted model updates.

Key management is a crucial issue in homomorphic encryption-based FL systems. The most common setting has all clients share a single private key $sk$, while the server only possesses the public key $pk$ [6], [23]. This single-secret-key approach is not only simple to implement, but it also fully protects client models in highly privacy-sensitive scenarios. Consequently, this cryptosystem is the most widely adopted and implemented in industrial applications, such as by IBM [25] and NVIDIA [26]. This paper also adopts the single-secret-key setting for its widespread use, efficiency, and strong privacy protection.

Some researchers also explore other settings like threshold HE [27] and multi-key HE [28]. Threshold HE [27] involves a group of users sharing different $sk$s, and a minimum number

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of users are required to collaborate to decrypt ciphertexts. It uses more complex cryptosystems like threshold Paillier. Furthermore, in Multi-key HE [28], decryption requires the collaboration of all clients. It uses a multi-key fully homomorphic encryption scheme MK-CKKS. However, these alternative settings incur much higher computational and communication costs compared to the single-secret-key scheme, making them very few applicable for industrial implementation. Therefore, this paper does not consider these settings.

### B. Backdoor Attacks and Defenses in FL

In federated learning systems, backdoor attacks pose a severe threat to the security and robustness of models [29], [30], [31]. These attacks are executed in a stealthy manner where malicious clients, either by poisoning their local datasets or directly manipulating model updates, introduce covert malicious functionality into the aggregated global model. Researchers have extensively studied this type of attack. Tolpegin et al. [32] investigate data poisoning, where adversaries introduce deleterious patterns into their local training data, subtly influencing the model to internalize malicious associations. Bagdasaryan et al. [33] propose weight tampering, where attackers deliberately scale their model updates to inject backdoors. When aggregated, such manipulations can compromise the global model. Xie et al. [7] introduce the distributed backdoor attack (DBA), where multiple malicious clients collaboratively inject parts of a global trigger pattern, enabling the backdoor to activate via the combined pattern. Backdoor attack is a serious problem in FL systems, as it can compromise the integrity and reliability of the shared model and lead to harmful predictions.

Various detection and mitigation techniques are proposed to defend backdoor attacks in FL systems. FLTrust [8] measures the trustworthiness of client models based on cosine similarity with a baseline model, which is trained by the server using its own clean dataset. However, FLTrust is incompatible with encrypted models in homomorphic encryption-based FL. Furthermore, FLCert [34] employs an ensemble learning strategy that aggregates multiple global models and uses majority voting for each prediction. However, it focuses on ensemble federated learning and cannot identify malicious clients attempting to poison the global model. Additionally, some recent work utilizes cryptography to address the backdoor threat. ShieldFL [13] employs a two-trapdoor homomorphic encryption scheme, where it measures encrypted abnormal gradients based on secure cosine similarity through the collaboration of two non-colluding servers. Similarly, Yazdinejad et al. [14] propose a framework that employs an internal auditor for malicious gradient detection through encrypted similarity analysis. However, both approaches face deployment challenges: ShieldFL requires two separate servers, while Yazdinejad's method relies on a trusted third-party auditor for key management and gradient verification.

Another line of work, including EIFFeL [15], RoFL [16], and ACORN [17], leverage zero-knowledge proofs (ZKP) to verify that local model updates satisfy certain constraints, such as bounds on $\ell_2$ or $\ell_\infty$ norm. These FL aggregation methods aim to mitigate the risk of poison attacks by filtering out illegal model updates. However, they face two key challenges: First, the cryptographic method ZKP typically incurs substantial communication and computational overhead. Second, while focused on satisfying the constraints, they lack a defense mechanism against Non-IID scenarios.

TABLE I
COMPARISON WITH EXISTING WORKS

| Method | Support HE | Supprot Non-IID | Server Number |
|---|---|---|---|
| FLTrust [8] | ○ | ● | 1 |
| FLCert [34] | ○ | ● | 1 |
| ShieldFL [13] | ● | ● | 2 |
| Yazdinejad et al. [14] | ● | ● | 2 |
| EIFFeL [15] | ● | ○ | 1 |
| RoFL [16] | ● | ○ | 1 |
| ACORN [17] | ● | ○ | 1 |
| Our HEFLGuard | ● | ● | 1 |

1 "●" and "○" indicate "yes", and "no" respectively.
2 ShieldFL requires two non-colluding servers. Yazdinejad et al.'s method requires a central server and a trusted internal auditor.

Table I provides a high-level comparison between HEFLGuard and representative defense mechanisms. As shown, methods such as FLTrust and FLCert are not designed for HEFL, while ShieldFL and Yazdinejad et al. rely on multiple non-colluding or trusted servers. ZKP-based schemes (EIFFeL, RoFL, ACORN) support encrypted settings but incur significant proof overheads and provide limited treatment of Non-IID and Byzantine behaviors. To our knowledge, HEFLGuard is the first framework that targets single-server HEFL, supports Non-IID settings, and avoids additional trusted parties while enabling effective backdoor and malicious-client detection with practical overhead.

## III. THREAT MODEL

We follow prior work [8], [33], [35] to set up our threat model. We consider a cross-silo HEFL setting with one central server and multiple clients. The system adopts the Paillier cryptosystem [5] in a single-secret-key setting (default key length 1024 bits [22]). The capabilities and knowledge of clients and servers are elaborated as below:

***Server***. The central server is *honest-but-curious*, tasked with the coordination of backdoor detection, including aggregating encrypted updates, distributing validation models, and analyzing client-reported statistics. The server follows the protocol faithfully, but attempts to learn as much as possible from received messages. It only holds the public key of HE and cannot decrypt client updates. Attacks that reveal the HE private key to the server, or fully compromise the server, are out of scope.

***Clients***. The clients can be represented as following:

• *Honest clients* follow the protocol without any malicious attempt: they train on their local data, evaluate the provided validation model on benign local validation samples, and return only the prescribed lightweight statistics. They never reveal plaintext model parameters or the HE private key to server, or collude with other clients.

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• *Malicious clients* may behave arbitrarily and collude: they can craft local updates to inject a targeted backdoor, tamper with their local validation data or falsify their reported statistics to compromise the integrity of aggregation.

***Adversary threshold***. Let $N$ denote the number of clients in a round and $m$ the number of malicious ones. HEFLGuard guarantees correct global backdoor diagnosis and malicious-client identification whenever the Byzantine fraction $\alpha = m/N$ satisfies

$$\alpha < \frac{\lfloor k/2 \rfloor + 1}{2k},$$

where $k$ is the validation group size. The derivation is detailed in Section V-E. For our default choice $k = 3$, this inequality simplifies to $\alpha < 1/3$. If this threshold is exceeded, the majority-consistency rule can be arbitrarily skewed; the TPR/FPR of the malicious clients identification may degrade significantly (see evaluation in Section VI-C.1). We therefore declare that the diagnosis is *high risk* and make no correctness guarantees for it.

***Realistic scenario***. In privacy-sensitive FL deployments, such as healthcare and finance, the coordinating server is usually a commercial cloud provider (e.g., AWS, Azure). The server operates under contractual and regulatory controls, and is thus assumed as honest-but-curious. Meanwhile, regulations such as HIPAA [36] motivate encrypting client model updates so the server only accesses encrypted models. In practice, most participating clients are honest while only minority may be compromised, which matches our majority-honest assumption.

## IV. Overview

### A. The Insight Behind Backdoor Detection

We introduce a novel backdoor detection approach that capitalizes on the prediction logits of models. In general, the task of a classification model is to ascertain the category of an input sample based on its features. However, the presence of backdoor can manipulate this task. The trigger can render the classification category independent of the features of the sample itself. We formalize the foundational concepts below.

*Definition 1 (Logits):* For an input $x$ and a model $M$, the logits produced by $M$ is the output of the last layer of the model, written as $\mathbf{y}_M(x) = \{y^1, y^2, \ldots, y^p\}$, where $p$ denotes the number of categories.

*Definition 2 (Logits Difference):* For two different models $M_1$ and $M_2$ of the same task, the logits difference for an input $x$ is defined as:$\mathbf{d}(x) = \mathbf{y}_{M_1}(x) - \mathbf{y}_{M_2}(x)$

Logits represent the model's confidence in each class based on the input's features. In a benign model (i.e. a model without backdoors), decision boundaries are formed based on the natural distribution of the training data. For models trained on independent and identically distributed (IID) data, their decision boundaries should align closely because they rely on the same underlying feature distributions. Hence, the logits differences between such models are small. When a backdoor is injected, the model is trained to associate a specific trigger with a target class, regardless of the actual features of the input sample. This results in an artificial decision boundary that significantly deviates from what would be learned from natural data distributions alone. The presence of a trigger causes the model to output logits that are surprisingly high for the target class and anomalously low for other classes, irrespective of the input features. Even when inferencing a benign input (i.e., a sample without the trigger pattern), the distortion in logits can still manifest itself because the model's skewed decision boundary already prioritizes the attack category over the normal category.

Below, we formally present two propositions:

*Proposition 1:* If two models $M_A$ and $M_B$ are trained on IID data for the same classification task, then for a benign input $x$, their output will lie close in the logits space. Formally, $|\mathbf{y}_{M_A}(x) - \mathbf{y}_{M_B}(x)| < \epsilon$, where $\epsilon$ is a small positive value.

*Proposition 2:* For a backdoored model $M_C$ targeting category $i$ and a benign model $M_B$, for a benign input $x$, the logits difference satisfies: $|y^i_{M_C}(x) - y^i_{M_B}(x)| > \Delta$. Here, $y^i$ denotes the logit for the $i$-th category, and $\Delta$ captures the significant distortion caused by the backdoor.

These propositions are empirically validated on multiple datasets and architectures in Section VI-B, where we show that backdoored models indeed exhibit characteristic shifts in logits distributions even on benign inputs. In Section V-D, we extend this logits-based consistency idea to Non-IID settings via a consistency-verification mechanism.

### B. Approach Overview

The insight illustrates that in a backdoored model, logits on benign samples are significantly large on the backdoor attack category compared to other non-backdoored models. To adapt homomorphic encryption-based FL, we introduce *validation models* as the object to compare with the global model. Validation models are constructed using aggregates from client subgroups. Inspired by cross-validation, we ensure that the groups of each validation model (i.e., validation groups) have some overlap, with each differing by one client. Consequently, any intersecting validation groups that detect the backdoor attack can effectively pinpoint the malicious clients. To optimize computational and communicative overheads, we propose to use only the last several layers (i.e., the classifier) of validation models, which can still maintain effectiveness.

In HEFL, the central server is constrained from performing inference computations on the model, but it can delegate these computational tasks to clients. The comparing models process is assigned to clients, who apply the models to their local test data and return the comparison results to the server. The server acts as a coordinator, facilitating the detection process. Through this collaborative effort between the server and clients, backdoor detection under homomorphic encryption can be effectively achieved. The symbols used in this paper are described in Table II. Fig. 1 shows the whole workflow of HEFLGuard. It consists of the following four steps:

1) **Train local model**: The client $C_i$ trains the initial model $M$ using its local dataset. Then $C_i$ encrypts its model parameters $M_i$ using the public key and uploads the model's ciphertext $E_{pk}(M_i)$ to the central server.
2) **Aggregate encrypted model**: The server divides clients' updates into $N$ overlapping groups, each consisting of $k$

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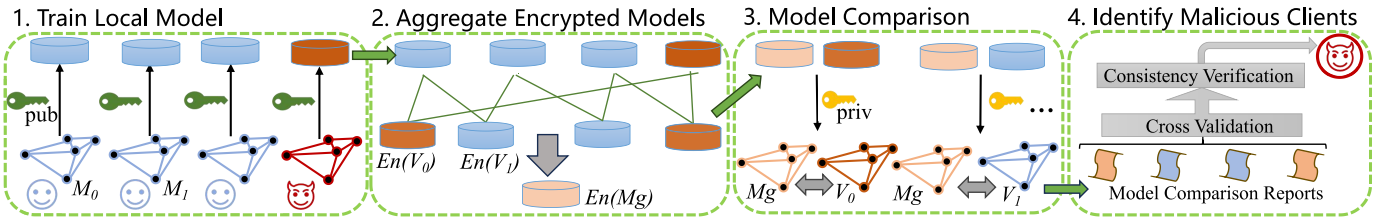



Fig. 1. HEFLGuard workflow. HEFLGuard performs a backdoor detection after each secure aggregation.

TABLE II
SYMBOLS USED IN THIS PAPER

| Symbol | Description |
|---|---|
| $N$ | The number of clients in FL system |
| $m$ | The number of malicious clients |
| $\alpha$ | The proportion of malicious clients |
| $C_i$ | The $i$-th client |
| $G_i$ | The $i$-th validation group |
| $k$ | The member count per validation group |
| $M_i$, $M_i^F$ | The model of client $C_i$ and its classifier |
| $M_{global}$, $M_{global}^F$ | The global model and its classifier |
| $V_i$, $V_i^F$ | The $i$-th validation model and its classifier |
| $E_{pk}(M_i)$ | The ciphertext of model $M_i$ |
| $Q$ | The validation dataset |
| $q$ | The number of validation samples each class |
| $x$ | The input sample |
| $f(x)$ | The feature vector of sample $x$ |
| $d(x)$ | The logits difference of sample $x$ |
| $z_i$ | The Z-score value for label $i$ |
| $T$ | The Z-score threshold for outlier detection |
| $\lambda$ | The scaling factor of the malicious updates |

models, and aggregates each group using HE to generate $N$ validation models. The server aggregates all models to obtain a global model. Then it distributes the global model and a random validation model to each client.

3) **Model Comparison**: Clients compute the logits difference of the global model and validation model on their local test samples, and report the results to the central server. The logits difference measures the inconsistency between the two models for each category, and a large difference indicates a potential backdoor attack.
4) **Identifying malicious clients**: The central server conducts statistics and cross-analysis of the detection results from all clients, and diagnoses whether the global model is backdoored, as well as predicts the target label of backdoor attack and identifies the malicious clients.

During the training process, the system repeats steps $1 \sim 4$. If the server confirms the same malicious client in multiple rounds of backdoor detection, it will remove the client from the system and aggregate the global model without malicious updates. Homomorphic encryption cryptosystem preserves the privacy of the model and enables secure aggregation, while the central server can still organize and perform backdoor detection with the encrypted model updates. Thus, HEFLGuard improves backdoor robustness and security of HEFL systems.

## V. DESIGN

This section first presents how HEFLGuard generates validation models and uses logits difference for efficient model comparison to detect backdoor. Then, we specify how the server groups clients and identifies malicious clients through cross validation of the comparison results. Furthermore, we demonstrate that HEFLGuard remains effective even in Non-IID data scenarios by employing a consistency verification scheme. Finally, we discuss that the server can utilize Byzantine fault-tolerant verification to get reliable detection conclusions even when malicious clients give dishonest results.

### A. Validation Models Construction

We consider a homomorphic encryption-base FL system with $N$ clients, denoted by $C_1, C_2, \ldots C_N$. Each client has a local model $M_i$. Among them, some are malicious and others are benign. To construct the validation models, we partition the clients into $N$ overlapping groups. Each group $G_i$ contains $k$ clients, and each client participates in $k$ groups. To preserve the privacy of clients, we set the group size $k$ is not less than 2. The central server aggregates the models of each group to produce a validation model $V_i$. Formally, the validation model can be computed as:

$$E_{pk}(V_i) = \frac{1}{k} \sum_{C_j \in G_i} E_{pk}(M_j) \tag{1}$$

$$G_i = \{C_i, C_{i+1}, \ldots, C_{(i+k-1) \bmod N}\} \tag{2}$$

In this way, we can generate $N$ overlapping validation models that can verify the results of each other.

To mitigate the computation and communication overhead, we introduce a strategy to use only the classifier layers of the validation model. A model can be partitioned into two discrete components: the feature extractor and the classifier. In general terms, the classifier operates closer to the output and adjudicates the relationships between the extracted features and their corresponding categories. We suggest that the central server only sends the classifier component during validation model transmission. Clients leverage the feature extractor from the global model for both models to get sample features, while employing their localized classifiers for logits comparison. During aggregation, the central server can then directly combine the client-side classifiers, denoted $M_F$, to form the validation model's classifier $V_F$. The classifier aggregation process is similar to the model aggregation process, which is formally written as:

$$E_{pk}(V_i^F) = \frac{1}{k} \sum_{C_j \in G_i} E_{pk}(M_j^F) \tag{3}$$

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In this way, we can generate $N$ validation classifiers. This technique can significantly reduce the additional communication cost caused by HEFLGuard without compromising its effectiveness. Moreover, limiting the transmission to only the classifier component minimizes the exposure of sensitive parameters in the validation model, thereby enhancing privacy safeguards. We further discuss about its effectiveness and overhead in Section VI-C3. And in Section VII-A, we discuss that when the attacker only attacks the feature extractor, he cannot successfully inject a backdoor into the global model, and it will greatly affect the performance of the main task.

### B. Client-Side Logits-Based Detection

As stated in our threat model, we assume that most clients are benign and only a few are malicious. When the server aggregates the encrypted models from the clients, it produces a global model and multiple validation models. We can reasonably assume that most of the validation models are benign. Then, the server sends the global model and a random validation model to each client respectively. At the client side, the client can use the validation model to perform backdoor detection for the global model.

Each client constructs a local validation set $Q$ exclusively from its private test split, with no training examples reused. We uniformly sample $q$ examples per class with default $q = 20$ to form $Q_c$. For honest clients, all examples in $Q$ are clean (without triggers). After each secure aggregation round, the server distributes the updated global model $M_g$ and one randomly selected validation model $V_i$ (classifier head only) to each client, which then performs the validation step on its fixed local set $Q$ before the next round begins. Malicious clients may tamper with their validation set or falsify reported results, such behavior is treated as part of the Byzantine setting and is analyzed in in Section V-E.

For each benign sample $x$, the client calculates its feature vector $f(x)$ using the feature extractor of the global model $M_{gloabl}$. Then, the client passes the extracted features $f(x)$ through both the classifier of the validation model $V^F$ and the classifier of the global model $M^F_{global}$, to obtain the output logits of the two models respectively. Then, for a given benign sample $x$, the logits difference $d(x)$ is calculated as:

$$d(x) = M^F_{global}(f(x)) - V^F(f(x)) \tag{4}$$

$d(x) = (d_1, d_2, \ldots, d_n)$ is an $n$-dimensional vector, with $d_i$ representing the logits difference of sample $x$ between the global and validation classifers on the category $i$.

We adopt the Modified Z-score algorithm [37] to perform outlier analysis on the logits difference. The Modified Z-score algorithm combines the advantages of the MAD and the Z-score methods. It measures the number of scaled absolute deviations of each data point from the median of the data set. The median is the value that divides the data set into two halves. The MAD is the median of the absolute deviations of each data point from the median. The modified Z-score algorithm is robust to outliers and does not assume any specific distribution of the data. A data point with a large modified Z-score value deviates significantly from the rest of the data set and is likely an outlier. Formally, the modified Z-score for the mean logits difference vector $\overline{d}$ can be calculated as:

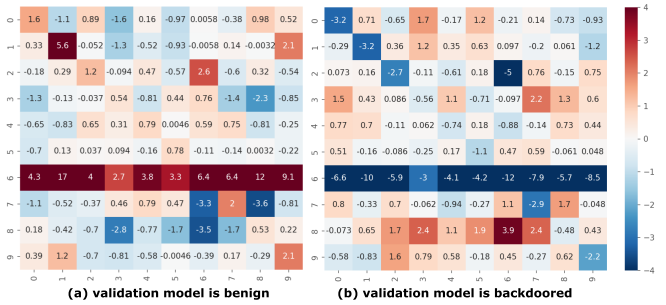



Fig. 2. Heatmaps of per-class $Z$-scores of logit differences $d(x) = M_{\text{global}}(f(x)) - V_i(f(x))$. The x-axis shows the ground-truth class; the y-axis shows the predicted class. (a): The global model is backdoored while the validation model is benign, and $Z$ concentrates positively on the target label "6". (b): the validation model is backdoored, and $Z$ concentrates negatively on the label "6"..

$$\overline{d} = \frac{1}{q}\sum_{x \in Q} d(x) \tag{5}$$

$$z_i = \frac{0.6745\overline{d_i} - \text{median}(\overline{d}))}{\text{MAD}(\overline{d})}, \quad i = 1, 2, \ldots, n \tag{6}$$

The constant 0.6745 [37] is used to scale the MAD so that it can be comparable to the standard deviation under a normal distribution.

To detect outliers, we define a threshold $T$, which is a parameter that determines the sensitivity of the outlier detection. In our case, we use $T = 3$ for it corresponds to the 99.7% confidence interval of a normal distribution [37]. A data point with a modified Z-score value greater than $T$ or less than $-T$ is an outlier. Formally, we can write:

$$\text{outlier}(\overline{d_i}) = \begin{cases} 1 & \text{if } z_i > T \\ -1 & \text{if } z_i < -T \\ 0 & \text{otherwise} \end{cases}, \quad i = 1, 2, \ldots, n \tag{7}$$

The algorithm detects whether the logits difference of benign samples between the global and validation classifiers has an outlier. If so, it indicates the global model has an abnormal classification bias not seen in the validation model. This strongly suggests the global model has been compromised by a backdoor attack, with the outlier likely being the target category of the attack.

We visualize the per-class Z-scores used for model comparison in Fig. 2. Each client computes class-level summary statistics and outlier flags locally and reports only a signed target to the server. No raw inputs, labels, or raw logits are transmitted. For clarity, we encode the decision using *signed targets*: "$+c$" if $z_c > T$ (the global model shows stronger bias toward label $c$ than the validation model) and "$-c$" if $z_c < -T$ (the validation model shows stronger bias).

For example, in Fig. 2(a), the global model is compromised by a backdoor attack, while the validation model is benign. The heatmap shows that the category corresponding to the backdoor target label "6" has a positive Z-score that exceeds the threshold $T$, so the client reports "+ 6" to the server,

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indicating that the global model is suspected to be backdoored targeting label "6". Conversely, in Fig. 2(b), the validation model is compromised, while the global model is less affected because it is aggregated from more clients. Here the Z-scores on label "6" are negative and below $-T$, so the client reports "−6", indicating that the validation model is suspected to be backdoored targeting label "6". This compact reporting yields negligible communication overhead while preserving privacy. Validation is processed after server-side aggregation and before the next round begins.

### C. Malicious Clients Identification

This section introduces the use of validation models to detect a single malicious client and extends the approach to handle multiple malicious clients. We first elaborate on the Independent and Identically Distributed (IID) data scenario and then extend the method to the Non-IID setting.

**Single Malicious Client**. Clients compare the validation model with the global model and obtain the outlier values outlier($\bar{d}$) for each category. A large absolute value of outlier($\bar{d}$) indicates a potential backdoor attack. Moreover, based on the sign of outlier($\bar{d}$), the central server can leverage cross-validation to identify the malicious client.

To identify the malicious client, the central server computes the intersection of the contributors of the backdoored validation models. The encrypted validation models and the global model are randomly distributed to clients. Each client decrypts the received models using their private key, conducts the model comparison, and reports the outlier values outlier($\bar{d}$) and the target category to the central server. Because clients do not know which of them participate in the received validation model, here we assume they report honest results (the dishonest scenario is discussed in Section V-E). If most clients report a backdoor attack on the global model with the same target category, the central server concludes that the global model is compromised with a backdoor. Based on the cross-validation principle, the only malicious client is the one that belongs to the validation groups whose models are reported to have a backdoor. Let $G_c$ be the validation groups that have negative outlier($\bar{d}$). Then, the malicious client $C_m$ can be identified as:

$$C_m = \bigcap G_c \tag{8}$$

**Multiple Malicious Clients**. In some cases, there may exists multiple malicious clients that participate in the HEFL system and try to inject backdoor into the global model. In our threat model, the malicious clients are controlled by the same attacker and use the same backdoor trigger and target class. To detect and identify these malicious clients, we need a more robust backdoor detection algorithm that can handle the collusion and consistency of the backdoor attacks.

We propose a skip selection strategy to deal with multiple malicious client scenarios, in which case benign clients that are surrounded by malicious ones can possibly become false positives. This strategy constructs two sets of validation models with different subsets of clients, and cross-validates them to identify the outliers. For example, suppose clients $C_2$ and $C_4$ are malicious, and $C_3$ is benign. If the defender selects clients only in the sequential order for constructing validation models, $C_3$ will be falsely classified as malicious. To avoid this situation, the defender can use skip selection to construct another set of validation models with a gap, such as 3. Formally, it can be written as:

$$G'_i = \{C_i, C_{i+3}, \ldots, C_{(i+3k-1) \bmod N}\} \tag{9}$$

By cross-validating two sets of validation models simultaneously, the server can significantly reduce false positives and false negatives in identifying malicious clients. While this skip selection strategy enhances the robustness of HEFLGuard, it also increases communication costs, as each client must perform backdoor detection on multiple validation models. Theoretically, using more validation sets could further improve accuracy, but it would significantly amplify communication overhead. However, our experiments show that using just two sets is already highly effective in handling multiple malicious clients. Moreover, the overhead remains manageable since only the classifier of each validation model is transmitted, which is much smaller than the entire model. Thus, our approach remains both efficient and practical.

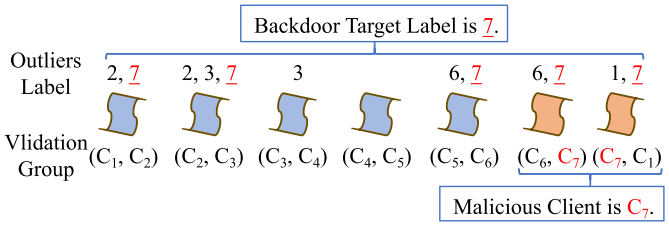



Fig. 3. Malicious clients identification in Non-IID case. We assume the client $C_i$ has a dominant category on label $i$. Malicious client $C_7$ injects a backdoor towards label 7.

### D. HEFLGuard in Non-IID Scenarios

In real-world scenarios, the data distributed among clients are probably non-independent and non-identically distributed. To handle the complex Non-IID cases in HEFL systems, we propose a consistency verification method to enhance backdoor detection. The main idea is to compare the outlier values outlier($\bar{d}$) of the logits difference between various validation models and the global model.

We consider the impact of Non-IID data from two perspectives: the global model and the validation models. For the global model, which is aggregated across diverse clients, the accumulated dataset tends toward class balance [1]. Such a balance ensures that a benign global model remains unbiased in its classifications. Hence, if the global model exhibit a conspicuously large outlier value for a specific category, it serves as a red flag, meaning that the category is a backdoor attack target. Conversely, the validation models usually have highly imbalanced data distributions, meaning that some categories have more samples than others in their local data. This can cause the benign validation models to have classification bias for their dominant categories, which are reflected in their outlier values. When analyzing the model comparison results, outliers include both dominant categories and the global model's backdoor attack category. Importantly,

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while a single client's dominant category might influence up to $k$ validation models, the global model's backdoor attack category permeates almost all validation model outcomes (as shown in Fig. 3). Consequently, if a majority of validation models pinpoint the same outlier category in their outliers, it's a clear marker of that category being the backdoor attack target. Therefore, our method is still effective and practical for the Non-IID scenarios.

We evaluate the effectiveness of HEFLGuard through experiments in Section VI-D.2. The results demonstrate that HEFLGuard can detect backdoor attacks and identify malicious clients with high accuracy in Non-IID settings.

### E. Byzantine Fault-Tolerance Verification

During backdoor detection process, malicious clients might alter reported results, aiming to mislead the central server's analysis, which is a common adversarial behavior known as Byzantine attack [38]. Specifically, these adversaries can tamper their local validation set or directly alter the comparison result to manipulate the result of malicious client identification. There are two situations if malicious clients change their model comparison results: (i) Concealing malicious behavior: A validation model might actually contain a backdoor, but malicious clients report otherwise. Their goal is to hide their involvement and elude detection. (ii) Creating false alarms: Conversely, a benign validation model is reported to have a backdoor. In this situation, the malicious clients aim to sow confusion, potentially leading the central server to wrongly flag benign clients. Both these behaviors threaten the integrity of the identification of malicious clients, which relies on consistency across validation models' reports to determine the presence of backdoor attacks and malicious clients.

However, there is a silver lining. Drawing from the principle of Byzantine fault-tolerance [39], as long as ratio of the malicious clients is within the adversary threshold, we can reach a consensus regarding the presence of a backdoor. We propose a Byzantine fault-tolerance verification algorithm based on the majority rule: if most validation models detect a similar outlier category, a backdoor attack is probable. As elaborated in Section III, malicious clients constitute only a minority (less than half). This ensures that any tampering by these clients will not significantly distort the overall backdoor detection results.

HEFLGuard leverages the overlapping nature of validation models: There are $N$ clients in total, and they are distributed in $N$ overlapping validation groups. Each validation group includes $k$ different client members, and each client is involved in $k$ different validation models. The clients send $N \cdot k$ reports overall. A group decision is made by *strict majority*; ties do not count as a flip.

Define the strict-majority threshold $\frac{\lfloor k/2 \rfloor + 1}{k}$. To flip any group outcome, at least $\lfloor k/2 \rfloor + 1$ Byzantine reports must occur within that group. Each Byzantine client can submit at most one false report per group and appears in exactly $k$ groups, so the adversaries collectively possess at most $m \cdot k$ false reports. Since each flipped group consumes at least $\lfloor k/2 \rfloor + 1$ such reports, the total number of flipped groups is upper-bounded by:

$$\#\text{flip} \leq \frac{m \cdot k}{\lfloor k/2 \rfloor + 1}$$

To remain the global majority correct, the flipped groups cannot exceed half of the total groups. Therefore, the upper bound of malicious clients $m$ should meet the requirement of the following formula:

$$\frac{m \cdot k}{\lfloor k/2 \rfloor + 1} < \frac{1}{2} \cdot N,$$

Equivalently, in terms of the Byzantine fraction $\alpha = m/N$,

$$\alpha < \frac{\lfloor k/2 \rfloor + 1}{2k}.$$

Under our random overlapping construction, such concentration is typically impossible, so this is a conservative guarantee. When $k = 3$, the condition reduces to $\alpha < 1/3$, thus even in the worst case, the attacker cannot flip half of the groups.

The overlapping validation system enhances the resilience of our consistency verification process against Byzantine reports. As long as the proportion of malicious clients remains within the effective boundary, even if the malicious clients alter their outlier categories, it will not affect the final detection result.

TABLE III
DATASET AND MODEL INFORMATION IN EXPERIMENTS

| Dataset | Model | Categories | Client Number | Trainset per Client |
|---|---|---|---|---|
| MNIST | LeNet5 | 10 | 50 | 1,200 |
| FASHION | LeNet5 | 10 | 20 | 3,000 |
| CHMNIST | AlexNet | 8 | 8 | 500 |
| CIFAR10 | VGG16 | 10 | 50 | 1,000 |
| CIFAR100 | ResNet34 | 100 | 10 | 5,000 |
| AG News | TinyBERT | 4 | 50 | 2,400 |
| Newsgroups | TinyBER | 20 | 10 | 1,130 |

## VI. EVALUATION

In this section, we first describe the experimental setup, then present the overall performance of HEFLGuard across seven datasets and three types of backdoor attacks under both IID and non-IID settings. Next, we conduct an ablation study on three key aspects of the design:validation group size $k$, the attacker's scaling factor $\lambda$, and classifier-only versus full-model validation. We further evaluate robustness under stronger adversaries, including multiple and distributed malicious clients as well as Byzantine behavior within the malicious clients. Finally, we compare HEFLGuard against prior defenses.

### A. Experimental Setup

The overall information of our experimental datasets and models is demonstrated in Table III.

***Datasets***: We conduct experiments on seven commonly used datasets: MNIST [40], FASHION [41], CHMNIST [42], CIFAR10 [43], CIFAR100 [43] AG News [44] and Newsgroups [45]. MNIST and FASHION each contains 60,000

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training and 10,000 testing $28 \times 28$ grayscale images of handwritten digits and clothing items respectively. CIFAR10 consists of 50,000 training and 10,000 testing $32 \times 32$ colored images of objects such as airplanes, birds, cats, and trucks. They are all 10-category classification tasks. CHMNIST (Colorectal Histology MNIST) comprises 5,000 $64 \times 64$ grayscale images of human colorectal cancer tissues, categorized into 8 distinct classes. The dataset is divided into 4,000 images for training and 1,000 images for testing. CIFAR100 extends CIFAR10 to 100 classes with 50,000 training and 10,000 testing $32 \times 32$ colored images. AG NEWS contains 120,000 training and 7,600 testing text samples from four news categories. The Newsgroups dataset contains 11,300 training and 7,500 testing text samples from 20 different newsgroup categories.

***Models***: For MNIST and FASHION, we employ LeNet5 [40], which is a well-established convolutional neural network. It consists of two convolutional layers with max-pooling, followed by three fully connected layers. For CIFAR10, we use VGG16 [46], which comprises 13 convolutional layers in five blocks, followed by three fully connected layers. For CHMNIST, we use AlexNet [47], which consists of five convolutional layers, followed by three fully connected layers with dropout. For CIFAR100, we use ResNet34 [48], which consists of 34 convolutional layers organized into residual blocks with skip connections. For AG News and Newsgroups, we employ TinyBERT [49], a compact transformer-based model distilled from BERT with 4 layers and 12 attention heads.

***Data distributions***: We investigate both IID and Non-IID setting. For the IID case, we randomly and non-overlappingly select the local datasets of clients from the overall training dataset, ensuring a roughly balanced distribution of each category across the clients. For the Non-IID case, we follow the mainstream method proposed in [35] to create data distributions with biases. Specifically, some clients have more data from a certain category than others. Parameter $p$ controls the Non-IID level. For example, if $p = 0.5$, then a sample with label $l$ has a 50% chance of being assigned to the $l$-th group, while other labels equally share the rest 50%. The higher the value of $p$, the more skewed the data distribution is.

***Backdoor types***: We evaluate four types of backdoors in different levels of difficulty and stealthiness: patch [50], blending [51], filter [52], and distributed backdoor attack (DBA) [7]. As presented in Fig. 4The patch backdoor is the simplest, adding a visible patch to the image. However, it can evade human detection if placed in a less noticeable area, such as a corner. The blending backdoor is more subtle, merging the trigger pattern with the input image to make it harder to detect. The filter backdoor is the most sophisticated, using color filters and image processing techniques to alter the image while keeping it natural, making detection difficult for both humans and machines. DBA is a type of backdoor attack targeted at federated learning systems. It involves the injection of a global trigger through the collaborative effort of multiple malicious clients, each injecting partial global trigger patterns. This distributed nature makes it significantly challenging to detect the backdoor and identify all malicious clients.

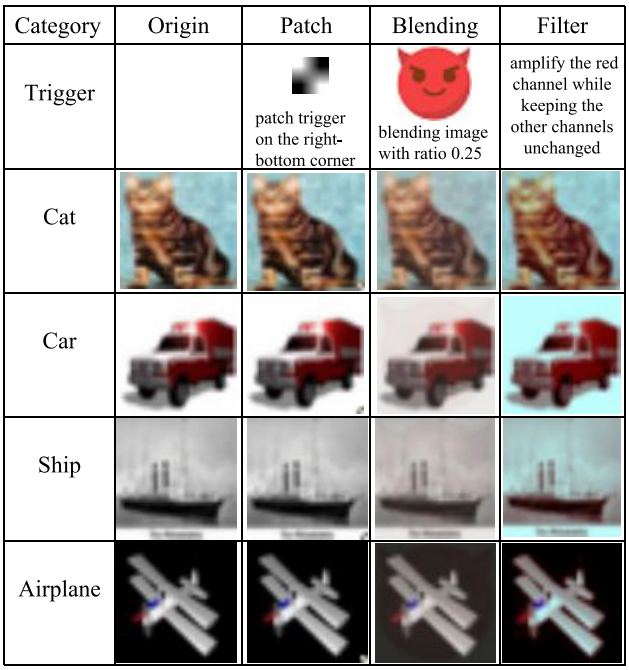



Fig. 4. Representative clean and poisoned inputs for different backdoor types on CIFAR-10. For each category, we show the original, patched, blended, and filtered image.

***Backdoored model training***: To simulate malicious clients, we follow the previous works [50], [33] and use a poison rate of $pr = 0.2$, modifying their local datasets with malicious samples. The backdoor is category-agnostic, meaning that it attacks all categories. Moreover, to inject backdoors effectively, we follow the mainstream method proposed in [33] to scale the gradients from the malicious models by a factor $\lambda$ before uploading them, i.e., $W_i = \lambda \cdot (M_i - M_{global})$, where $W_i$ is the parameter of client $C_i$'s local model and $W_g$ is the parameter of the global model. The choice of $\lambda$ depends on the number of clients and the trade-off between backdoor success and accuracy loss. A small $\lambda$ will make the backdoor injection fail, and a large $\lambda$ will degrade the main task accuracy.

***Homomorphic encryption setting***: The implementation of HEFLGuard is not tied to a specific homomorphic encryption cryptosystem, allowing it to be compatible with any HE scheme. In our experiments, we adopt the Paillier encryption scheme [5], [6], a widely applied cryptographic scheme that offers an additive homomorphism property, with a default key length of 1024 bits. For most applications, a key length of 1024 bits is generally considered secure [5]. This choice balances security and computational efficiency, aligning with typical requirements in FL scenarios.

***Evaluation metrics***: We evaluate the main task accuracy (ACC) and backdoor attack success rate (ASR) in the context of malicious homomorphic encryption-based FL systems. Additionally, we assess HEFLGuard's accuracy in locating malicious clients that perform backdoor injection. To measure the malicious client identification (MCI) performance, we employ the true positive rate (TPR) and false positive rate (FPR). TPR represents the rate of correctly identified malicious clients, while FPR indicates the rate of benign clients incorrectly classified as malicious.

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TABLE IV
FEDERATED LEARNING WITH HEFLGUARD RESULTS OF SEVEN DIFFERENT DATASETS WITH THREE DIFFERENT BACKDOORS

| Setting | Dataset | No backdoor | | Patch | | | | Blending | | | | Filter | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ACC | ASR | ACC | ASR | TPR | FPR | ACC | ASR | TPR | FPR | ACC | ASR | TPR | FPR |
| **IID** | MNIST | 98.7 | 10.4 | 98.4 | 11.9 | **100.0** | **0.0** | 98.7 | 13.6 | **100.0** | **0.0** | 98.1 | 12.6 | **100.0** | **0.0** |
| | FASHION | 87.1 | 11.1 | 86.8 | 11.6 | **100.0** | **0.0** | 85.8 | 9.9 | **100.0** | **0.0** | 85.7 | 10.9 | **100.0** | **0.0** |
| | CHMNIST | 67.2 | 12.2 | 67.1 | 12.7 | **100.0** | **0.0** | 66.7 | 9.5 | **100.0** | **0.0** | 67.3 | 13.2 | **100.0** | **0.0** |
| | CIFAR10 | 86.3 | 10.2 | 86.1 | 10.2 | **100.0** | **0.0** | 86.2 | 11.4 | **100.0** | **0.0** | 85.9 | 9.7 | **100.0** | **0.0** |
| | CIFAR100 | 59.7 | 1.8 | 58.5 | 1.6 | **100.0** | **0.0** | 57.9 | 1.9 | **100.0** | **12.5** | 57.4 | 2.0 | **100.0** | **12.5** |
| | AG News | 92.5 | 23.4 | 92.4 | 23.4 | **100.0** | **0.0** | - | - | - | - | - | - | - | - |
| | Newsgroups | 62.7 | 3.08 | 62.3 | 2.51 | **100.0** | **0.0** | - | - | - | - | - | - | - | - |
| **Non-IID** | MNIST | 98.6 | 10.7 | 98.2 | 12.3 | **100.0** | **0.0** | 98.3 | 9.6 | **100.0** | **0.0** | 98.4 | 13.6 | **100.0** | **0.0** |
| | FASHION | 86.9 | 12.1 | 86.9 | 11.3 | **100.0** | **0.0** | 85.9 | 12.1 | **100.0** | **0.0** | 86.1 | 9.8 | **100.0** | **0.0** |
| | CHMNIST | 66.5 | 13.9 | 64.7 | 16.5 | **100.0** | **0.0** | 66.1 | 15.9 | **100.0** | **0.0** | 67.1 | 12.8 | **100.0** | **0.0** |
| | CIFAR10 | 85.2 | 10.1 | 84.9 | 10.4 | **100.0** | **0.0** | 84.8 | 11.3 | **100.0** | **0.0** | 84.9 | 10.8 | **100.0** | **0.0** |
| | CIFAR100 | 59.1 | 1.9 | 58.4 | 1.8 | **100.0** | **0.0** | 57.5 | 1.9 | **100.0** | **12.5** | 57.4 | 2.1 | **100.0** | **12.5** |
| | AG News | 91.2 | 23.7 | 90.9 | 24.8 | **100.0** | **0.0** | - | - | - | - | - | - | - | - |
| | Newsgroups | 62.2 | 2.62 | 60.7 | 3.15 | **100.0** | **0.0** | - | - | - | - | - | - | - | - |

[1] The TPR, FPR (%) represent the accuracy of the malicious client identification (MCI) metrics in this paper.
[2] For the AG News and Newsgroups text classification task, we only evaluate the patch-type backdoor attack.

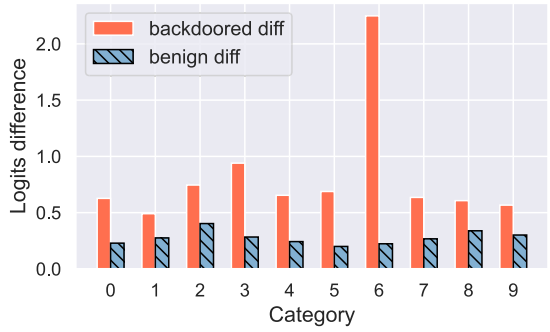



Fig. 5. Logits differences among benign models ($M_A$, $M_B$) and a backdoored model ($M_C$) for VGG16 on CIFAR10. The red bar shows the mean logits difference between $M_C$ and $M_A$, $M_B$, while the hatched blue bar denotes the logits difference between $M_A$ and $M_B$. $M_C$ is a backdoored model with attack class 6.

### B. Overall Performance

To empirically validate the propositions in Section IV-A, we train two benign models ($M_A$, $M_B$) and one backdoored model ($M_C$) using VGG16 on CIFAR10. We randomly select 200 benign test samples and compute their logits under all three models. As shown in Fig. 5, even without triggers, the backdoored model exhibits a distinctly anomalous logits distribution in the target category compared to the two benign models. For benign models, the logits differences remain consistent across a wide range of inputs, whereas the skewed decision boundary of the backdoored model introduces a systematic bias in logits that can be statistically distinguished from the natural variability among benign models. This observation supports our core intuition: backdoor effects are typically accompanied by characteristic shifts in logits distributions on benign inputs.

We assess the effectiveness of HEFLGuard in detecting patch, blending, and filter backdoors under both IID and Non-IID scenarios. Table IV summarizes the final main-task accuracy (ACC), backdoor attack success rate (ASR), and malicious-client identification (MCI, reported as TPR/FPR) across seven datasets. Varying the attack target and the malicious client, HEFLGuard consistently attains 100% TPR across all five vision tasks as well as the text classification task. On CIFAR100 we observe a small but non-zero FPR (12.5%) for blending and filter backdoors, whereas all other datasets achieve 0% FPR. We attribute this to the intrinsic difficulty of the 100-class task and the resulting variance among benign clients. Because each client holds only limited training data for each class, some benign clients naturally produce gradients that deviate more strongly from the global updates. Even in this case the final main-task accuracy on CIFAR100 only drops by about 2% points compared to the no-backdoor baseline, indicating that the impact of this FPR is limited. We also experiment with 50 clients on CIFAR100 to match the MNIST setting. However, this substantially reduces the per-client data available for the 100-class task (1,000 images per client), yielding a low-utility base model (e.g., vanilla FedAvg ACC decreases from 59.7% with 10 clients to 39.8% with 50 clients). In such a low-utility regime, per-class prediction confidence statistics become highly noisy and exhibit large variance, which undermines the stability of the logits difference signals used by HEFLGuard. Therefore, we report CIFAR100 with 10 clients as the main evaluation setting, and explicitly note that HEFLGuard assumes a base model with reasonable main-task utility for the detection statistics to be well-behaved.

We note that the "No-backdoor" result in Table IV is already the matched federated baseline under the same client partition and training protocol, rather than a centralized-training upper bound. Therefore, the relevant utility overhead of HEFLGuard is the additional accuracy drop relative to this matched FL baseline, which remains modest. This small gap is also consistent with our mitigation strategy: once a client is repeatedly identified as malicious, its update is excluded from subsequent aggregation to prevent poisoned information from further contaminating the global model, so any potentially useful task-related contribution contained in that suspicious update is also discarded. This effect is slightly more visible on CIFAR100, where the benign inter-client variance is larger and the detection statistics are less stable, as explained above.

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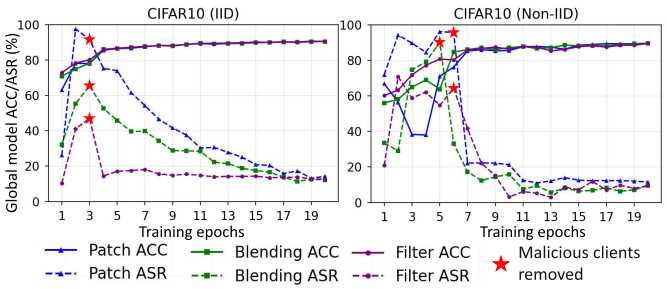



Fig. 6. The main task accuracy (ACC %) and backdoor attack success rate (ASR %) of HEFLGuard on CIFAR10 datasets with three types of backdoors in IID and Non-IID scenarios.

Fig 6 illustrates the evolution of ACC and ASR on the CIFAR10 task as a representative example. In this setting, HEFLGuard typically detects malicious clients within the first few epochs (e.g., within five epochs in the CIFAR10 task) and removes them from the FL system. After removal, the remaining benign clients continue training until convergence and the backdoor effect gradually vanishes. In Non-IID settings, the heterogeneous data distribution introduces additional challenges. Detection usually occurs 1–3 epochs later than in the IID case, and the corresponding federated training also converges more slowly. Nevertheless, HEFLGuard still achieves 100% TPR and 0% FPR for identifying malicious clients under the moderate Non-IID level ($p = 0.5$).

### C. Ablation Study: Hyperparameters and Design Choices

We perform an ablation study on three key knobs in HEFLGuard: (i) the validation group size $k$, (ii) the gradient scaling factor $\lambda$ used by attackers, and (iii) the design choice of sending the whole model versus only the classifier for validation. We report the impact on both clean accuracy and backdoor behavior.

*1) Effect of Validation Group Size:* We first ablate the validation group size $k$, which controls how many clients jointly contribute to each validation model. Table V reports the malicious-client identification performance (TPR/FPR) for different $k$ on MNIST and CIFAR-10. When the fraction of malicious clients is moderate and well below the theoretical tolerance bound, the choice of $k$ has almost no impact on MCI performance: all configurations achieve (100%) TPR and essentially zero FPR across $k \in \{2, 3, 4, 5\}$. In these settings we also verified that the global model's clean accuracy and ASR remain almost unchanged when varying $k$, staying close to the values reported in Tab. IV. This indicates that the validation group size primarily affects the reliability of MCI rather than the underlying task performance.

Differences become more apparent as the malicious-client ratio approaches the theoretical $\alpha$-bound for each $k$. For MNIST with 30% malicious clients, MCI remains strong with (100%) TPR and FPR $\leq 4.0\%$ for $k \leq 4$. However, for $k = 5$, where the malicious ratio slightly exceeds the $\alpha$-bound, TPR drops to 78.7% and FPR increases to 9.71%. A similar trend is observed on CIFAR-10: TPR stays at 100% for $k = 2$ and $k = 3$, but decreases to 66.7% and 46.7% for $k = 4$ and $k = 5$, respectively. These results show that overly large validation

TABLE V
HEFLGUARD PERFORMANCE UNDER DIFFERENT VALIDATION GROUP SIZE AND MALICIOUS CLIENTS RATIO

| Dataset | Malicious Clients | Validation group size $k$ | | | |
|---|---|---|---|---|---|
| | | $k = 2$ | $k = 3$ | $k = 4$ | $k = 5$ |
| $\alpha$ bound | | 0.50 | 0.333 | 0.375 | 0.30 |
| MNIST | 5/50 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 |
| | 10/50 | 100.0/0.0 | 100.0/1.0 | 100.0/0.0 | 96.0/6.00 |
| | 15/50 | 100.0/0.0 | 100.0/4.0 | 100.0/2.86 | 78.7/9.71 |
| | 20/50 | 100.0/0.0 | 97.0/11.3 | 95.0/10.0 | 45.0/2.67 |
| CIFAR10 | 1/10 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 |
| | 2/10 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 |
| | 3/10 | 100.0/0.0 | 100.0/0.0 | 66.7/2.85 | 46.7/2.85 |
| | 4/10 | 100.0/10.0 | 80.0/13.3 | 10.0/6.67 | 20.0/6.67 |

[1] Each cell reports the malicious-client identification (MCI) (TPR/FPR, in %).
[2] The '$\alpha$-bound' gives the theoretical upper bound on the tolerated fraction of malicious clients for each $k$ (see Sec. III).
[3] Entries highlighted as '*' correspond to settings where the actual malicious client ratio exceeds the $\alpha$-bound.

groups become brittle when the malicious ratio is close to the theoretical limit, because each validation model is more likely to be contaminated by multiple adversaries.

Overall, the ablation in Tab. V suggests that $k = 3$ achieves a good trade-off: it offers a strict theoretical adversary threshold $\alpha = 1/3$ while maintaining consistently high MCI TPR and low FPR across all tested settings where the malicious ratio does not exceed the $\alpha$-bound. Consequently, we adopt $k = 3$ as the default in all other experiments.

*2) Effect of Scaling Factor \lambda:* In our experiments, we use scaling factor $\lambda$ to control the strength of the backdoor attack when malicious clients update their poisoned gradients, i.e., $W_i = \lambda \cdot (M_i - M_{global})$. Larger $\lambda$ values make the adversarial clients push the backdoor objective more aggressively during aggregation, while smaller $\lambda$ lead to much weaker poisoning signals. Table VI reports HEFLGuard's performance for different $\lambda$ under a fixed 20% fraction of malicious clients with blending type backdoors. For MNIST (10/50 malicious clients) and CIFAR10 (2/10 malicious clients), HEFLGuard achieves 100.0% TPR and 0.0% FPR when identifying malicious clients for $\lambda \geq 3$. Though HEFLGuard misses a small fraction of such extremely weak attackers for $\lambda = 2$, we note that $\lambda = 2$ corresponds to very weak malicious gradients. Thus, the residual undetected backdoor has negligible practical impact, the poisoning barely changes the global model and the final backdoor attack success rate remains close to that of a benign model (below 15%). This ablation shows that HEFLGuard remains effective across a wide range of malicious-gradient strengths and does not depend on unrealistically aggressive backdoor updates.

*3) Classifier-Only Vs Full-Model Validation:* As discussed in Section V-B, HEFLGuard does not require the entire model to be transmitted as an encrypted validation model. Instead, we can keep the feature extractor on the client side and only homomorphically aggregate the classifier portion of the model. Each client uses the global model's feature extractor to obtain intermediate representations and then feeds these features into the classifiers of both the global and validation models. By comparing the logits between the two classifiers

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TABLE VI

HEFLGUARD PERFORMANCE UNDER DIFFERENT SCALING FACTOR $\lambda$ WITH 20% MALICIOUS CLIENTS AND BLENDING BACKDOOR

| Dataset | Malicious Clients | Scaling factor $\lambda$ | | | |
|---|---|---|---|---|---|
| | | $\lambda = 2$ | $\lambda = 3$ | $\lambda = 4$ | $\lambda = 5$ |
| MNIST | 10/50 | 90.0/0.0 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 |
| CIFAR10 | 2/10 | 80.0/0.0 | 100.0/0.0 | 100.0/0.0 | 100.0/0.0 |

TABLE VII

OVERHEAD AND HEFLGUARD DETECTION RESULTS

| Validation model | Entire model | 3FC | 2FC | 1FC |
|---|---|---|---|---|
| Parameter amount | 44,426 | 41,854 | 11,014 | 850 |
| Encryption time(s) | 15.21 | 10.79 | 5.45 | 2.26 |
| Decryption time(s) | 12.46 | 8.34 | 5.02 | 2.13 |
| Aggregation time(s) | 3.07 | 2.50 | 1.32 | 0.59 |
| Model comparison(s) | 0.42 | 0.25 | 0.22 | 0.21 |
| Ciphertext size(KB) | 12,799 | 12,057 | 3,173 | 246 |
| OC server(s) | 35.10 | 33.44 | **9.41** | **0.74** |
| OC client(s) | 174.6 | 160.2 | **43.6** | **2.34** |
| Overhead traffic(%) | 100.0 | 94.2 | **24.8** | **2.1** |
| HEFLGuard MCI | 100/0.0 | 100/0.0 | 100/0.0 | 100/0.0 |
| ACC/ASR | 86.5/11.9 | 86.5/11.9 | 86.8/11.6 | 86.6/11.1 |

[1] Aggregation time refers to time cost by aggregating one validation model.
[2] OC server refers to the overhead computation time of server to aggregate validation models.
[3] OC client refers to the overhead computation time of client to decrypt a validation model and model comparison.
[4] Overhead traffic refers to transferring validation models.

in the clear, HEFLGuard can still identify malicious clients while potentially saving a substantial amount of computation and communication.

To quantify this design choice, we consider a HEFL system that trains a LeNet5 model on the FASHION dataset with 20 clients. LeNet5 consists of two convolutional layers followed by three fully connected layers. We instantiate four validation-model configurations: (i) the entire model, and (ii) only the last three fully connected layers (3FC), (iii) only the last two fully connected layers (2FC), and (iv) only the last fully connected layer (1FC). Table VII reports MCI performance for these four configurations in the the last row. We see that HEFLGuard achieves the same malicious-client identification performance (100/0.0) across all four configurations. We further verified that the global clean accuracy and ASR at convergence remain identical up to small fluctuations, meaning that restricting the validation model to the classifier does not degrade either clean performance or backdoor mitigation effectiveness.

On the other hand, the overhead differs dramatically. Moving from the entire model to the 2FC configuration reduces the ciphertext size from 12,799 KB to 3,173 KB and the relative validation traffic to only 24.8% of the full-model baseline, while the 1FC configuration further shrinks the ciphertext size to 246 KB and the validation traffic to merely 2.1%. A similar trend holds for encryption and decryption: the client-side overhead for validation (OC client) drops from 174.6 s to 43.6 s (2FC) and 2.34 s (1FC), and the server-side overhead (OC server) falls from 35.10 s to 9.41 s and 0.74 s, respectively. The aggregation and model-comparison times are negligible compared to encryption and decryption.

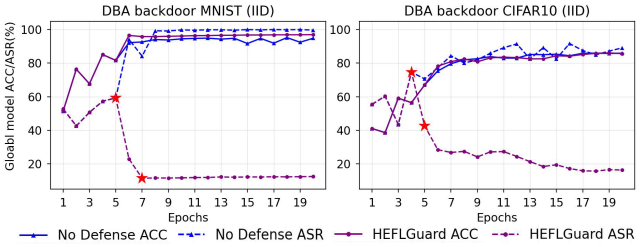



Fig. 7. HEFLGuard against DBA backdoor.

These results show that HEFLGuard can safely replace full-model validation with a classifier-only validation model, achieving the same detection performance at a fraction of the computational and communication cost. Although the 1FC configuration can reduce overhead to the minimum, it also exposes a larger attack surface for adaptive adversaries: by freezing the last fully connected layer and training the other layers, attackers may also inject backdoors (Sec. VII-A considers this adaptive attack). Therefore, we adopt the 2FC configuration as the default, which offers a balance between efficiency and robustness against such adaptive attacks.

### D. Robustness Under Stronger Adversaries

*1) Multiple and Distributed Malicious Clients:* With the validation group size fixed to $k = 3$, we now vary the number of malicious clients to characterize how HEFLGuard behaves as the adversary population grows. The $k = 3$ column of Tab. V summarizes the resulting MCI TPR/FPR on MNIST and CIFAR-10. As long as the malicious-client ratio stays below the adversary threshold $\alpha$-bound, HEFLGuard maintains near-perfect MCI performance. In all these cases, the global model's clean accuracy and ASR remain close to the no-defense HEFL baseline once detected malicious clients are removed. However, when the malicious ratio exceeds the $\alpha$-bound, MCI performance degrades. For MNIST with 20/50 malicious clients, where the ratio violates the $\alpha$-bound for $k = 3$, the MCI TPR remains high at 97.0% with FPR increasing to 11.3%. Similarly, on CIFAR-10 with 4/10 malicious clients, HEFLGuard still identifies 80.0% of malicious clients at the cost of a higher FPR of 13.3%. These results indicate that, even beyond the adversary threshold, the system continues to remove the majority of attackers, albeit with an increased fraction of honest clients being misclassified.

Furthermore, we evaluate HEFLGuard against the DBA (Distributed Backdoor Attack) [7], where four malicious clients inject distinct segments of a global trigger. As depicted in Fig. 7, HEFLGuard successfully identifies all four malicious clients. Specifically, in the CIFAR10 task, HEFLGuard firstly identifies and removes two of the malicious clients, causing the global trigger to lose its effectiveness. Then, HEFLGuard detects the remaining two malicious clients. This phased mitigation is consistent with the MNIST case, where HEFLGuard initially detects three malicious clients and later identifies the last one after a few epochs. This demonstrates the effectiveness of our approach in dealing with multiple malicious clients and distributed backdoors.

*2) Robustness to Cheating Clients:* We assess the effectiveness of Byzantine fault-tolerance verification of HEFLGuard,

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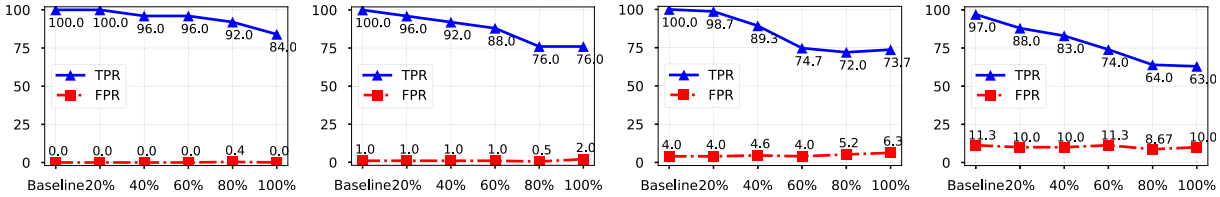



Fig. 8. TPR and FPR(%) of MCI results on MNIST dataset with different proportions of dishonest malicious clients. Subtitles are (Number of malicious clients)/(Total number of clients). X-axis is the percentage of dishonest malicious clients.

in case malicious clients submit deceptive results, i.e., a benign validation model is reported as backdoored, and a backdoored validation model is reported as benign. We focus on two aspects: global model diagnosis and malicious client identification. Impressively, under different levels of dishonesty among the malicious clients, the diagnosis of the existing of backdoor in global model remains 100% TRR and 0% FPR consistently. This outcome aligns with our expectation that a backdoor can be identified as long as a majority of clients are honest. However, MCI's accuracy declines as the number of dishonest malicious clients increases, as shown in Fig. 8. When the number of malicious clients is small, MCI reports few false positives, as in Fig. 8 (a) (b). As the proportion of malicious clients rises, their dishonest reports begin to impact MCI's accuracy, as in Fig. 8 (c) (d). Despite this, HEFLGuard maintains a TPR exceeding 70% and a FPR around 10% or lower in most cases of dishonest clients, demonstrating the robustness of our approach.

### E. Comparative Analysis With Prior Defenses

In this subsection, we provide a comparative analysis that benchmarks HEFLGuard against several representative defenses, including Trim-mean [18], Krum [19], FLTrust [8] and RoFL [16]. Our evaluation encompasses patch, blending, and Filter backdoors across various attack scenarios. Table IV has already quantified HEFLGuard's utility–security trade-off, HEFLGuard attains comparable performance while achieving the low ASR and modest clean-accuracy degradation with the no-backdoor setting reported in Table IV. For the other defenses, prior defenses can provide a certain level of protection for simple patch and blending backdoors and single malicious client scenarios, obtaining similar ASR and clean accuracy as HEFLGuard. However, for stealthy Filter backdoors with multiple malicious clients, the gap becomes pronounced. As shown in Fig. 9 for the FASHION dataset with four malicious clients, HEFLGuard successfully identifies all malicious clients without false positives across multiple consecutive epochs and reduces the backdoor ASR to around 10% at convergence, while maintaining competitive clean accuracy. In contrast, Trim-mean, RoFL and FLTrust show almost no resistance, with ASR values approaching 100%, and Krum exhibits unstable behavior with ASR fluctuating between 10% and 40%.

Trim-mean, Krum and FLTrust rely on direct examination of plaintext model updates, which is incompatible with homomorphic-encryption-based FL. Although RoFL employs zero-knowledge proof protocols to filter abnormal updates under HEFL, it still fails to mitigate sophisticated stealthy

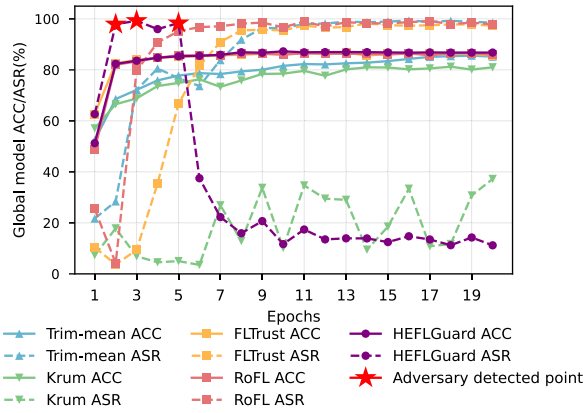



Fig. 9. HEFLGuard compares with Trim-mean [18], Krum [19], FLTrust [8] and RoFL [16] against stealthy backdoor on FASHION dataset.

backdoor attacks in our experiments. Consequently, HEFLGuard is a practical defense that addresses the security gap in encrypted FL systems by eliminating stealthy backdoors without compromising privacy guarantees or detection reliability.

We also measure the end-to-end round cost of HEFLGuard and compare it with RoFL [16]. On a LeNet5 model with approximately 62k parameters and 48 clients on the CIFAR10 dataset, HEFLGuard takes 88 s per round, and each client sends 19.9 MB per round. By contrast, RoFL's $\ell_2$ defence reports 110 s per round and 14.2 MB per client in the same configuration. Its client-side complexity scales as $O(\ell \log n)$ for computation and $O(\ell + \log n)$ for communication, which implies that both runtime and communication costs grow faster as the number of clients $n$ increases. In our HEFL setting, HEFLGuard keeps both computation and communication linear in the model size and achieves smaller time overhead, making it practical and scalable for realistic HEFL systems.

## VII. Discussion

### A. Adaptive Attackers

Since HEFLGuard usually detects backdoors based on the fully connected layers, we consider adaptive attackers who attempt to evade detection by utilizing a benign model classifier. To achieve this, the attackers can freeze the fully connected layers of the model and train only the convolutional layers to inject backdoors. We test this strategy using the FASHION dataset and a LeNet5 model with 20 clients, where one client is malicious and attempted to inject a backdoor with a patch trigger. As a control, when the fully connected layers are not frozen, the malicious client multiplies the

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local model parameters by a default value of $\lambda = 16$ and uploads them to the central server, achieving an ASR of more than 90% on the global model. In contrast, when the fully connected layers were frozen, the ASR for the global model decreased to less than 30% for $\lambda = 16$. Attempts to increase $\lambda$ to 24 resulted in a collapse of the main task, with test accuracy fluctuating between 20% and 60%. We also investigated attackers' use of regularization to evade detection. Our experiments employed $L_2$ regularization, which constrains the Euclidean norm of the weight parameters in the updated model. Results indicate that regularized models struggle to inject effective backdoors (ASR< 20%), while HEFLGuard maintained its ability to detect malicious clients. These findings underscore the inherent trade-off malicious participants face between evading detection and achieving a high ASR.

### B. Privacy Considerations

In federated learning, participation information leakage can lead to *Membership Inference Attacks* (MIA) [4], where an adversary attempts to determine whether a given input $x$ belongs to a specific client's training data. The inference function $I(M, x)$ assesses membership based on a queried model $M$. We consider a HEFLGuard system, where a central server coordinates $n$ clients, among which $m$ may be malicious.

**MIA via the Global Model**. When an adversary queries the global model $M_g$, they can infer whether $x$ was used in the entire FL system but cannot pinpoint the specific client. Since the adversary excludes themselves (malicious clients), the probability of correctly attributing $x$ to the true client is:

$$P_{\text{MIA}}\left(M_g\right) = \frac{1}{n-m}$$

**MIA via the Validation Model**. Each validation classifier $V_{Fi}$ is randomly constructed from $k$ clients before distribution. To infer membership from $V_{Fi}$, an adversary must first determine which $k$ clients contributed to it and then identify the owner of $x$. Given that $t$ of these $k$ clients are malicious ($0 \le t \le m, t < k$), the probability of correctly attributing $x$ can be computed. First, the adversary must infer the correct subset of $k$ clients: $P_{\text{subset}} = \frac{C_{n-m-1}^{k-t-1}}{C_{n-m}^{k-t}} = \frac{k-t}{n-m}$. Secondly, identifying the correct client within that subset: $P_{\text{client}} = \frac{1}{k-t}$. Thus, the overall probability of a successful inference is:

$$P_{\text{MIA}}\left(V_{Fi}\right) = P_{\text{subset}} \cdot P_{\text{client}} = \frac{k-t}{n-m} \cdot \frac{1}{k-t} = \frac{1}{n-m}$$

Since $P_{\text{MIA}}(V_{Fi}) = P_{\text{MIA}}(M_g)$, inference through validation classifiers provides no additional advantage over the global model. Therefore, HEFLGuard does not increase privacy risks beyond the inherent MIA vulnerabilities in federated learning.

We further quantify the potential privacy leakage using an empirical membership-inference attack. We consider an adversary that, given a trained model and an input $x$, predicts whether $x$ was used during training based on the model's prediction confidence. We evaluate this attack on the global model and on per-client models in our federated setting on the CIFAR-10 task. For the global HEFL model, the attack achieves an accuracy of 59.33%, compared to 50% for random guessing, corresponding to a small inference advantage of 9.33%. For a specific client model, where non-member samples are drawn from other clients' training data that still contribute to the global model, the attack accuracy is 50.11%, i.e., essentially random. These measurements indicate that: (i) HEFLGuard's use of validation classifiers does not introduce additional membership leakage beyond that of the underlying global model, and (ii) inferring the originating client of a training sample remains hard in practice, even with full access to that client's final local model parameters. Thus, under this attack model, the additional privacy risk introduced by HEFLGuard with respect to membership inference is limited.

## VIII. Conclusion and Future Work

We presented HEFLGuard, a backdoor detection framework for homomorphic encryption-based federated learning, showing that encrypted aggregation and backdoor auditing are compatible under a standard single-server setting. Through our extensive empirical evaluation, HEFLGuard shows that the server-side structuring of encrypted updates combined with lightweight client-side logits feedback, can expose backdoor behaviors while preserving homomorphic security guarantees. This insight suggests a general design pattern for future secure FL protocols that must operate under encryption and limited observability.

At the same time, several limitations open up directions for further work. HEFLGuard assumes a single-secret-key HEFL deployment with an honest-majority of clients and sufficient benign local validation data. Future extensions include supporting multi-key and threshold HE settings, scaling to highly heterogeneous models and devices, and designing defenses against more adaptive or stealthy attackers. Another promising avenue is integrating HEFLGuard with compression and protocol-level optimizations to further reduce its overhead, as well as combining it with complementary verifiable or certified defenses. We hope that HEFLGuard can serve as a foundation for practical, privacy-preserving, and attack-aware federated learning in real-world encrypted deployments.

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